Practical - 1

Practical Title: Project Initiation, Requirement Analysis, Sitemap, Wireframe, and GitHub Setup

Problem Definition: Initiate the semester-long StudentHub portal by identifying the problem scope, user roles, key modules, navigation flow, and minimum 10 pages. Create a sitemap, low-fidelity wireframe, project folder structure, README file, and GitHub repository.

A. Theory & Analysis

1. What is URL and parts of URL?

A URL (Uniform Resource Locator) is the address used to access web resources. Its primary parts are: (1) Protocol: Defines the transfer method, such as HTTP or HTTPS; (2) Subdomain: Directs requests to specific sub-services; (3) Domain: The main host name of the university or website; (4) Port: The communication port (such as :8000 for local PHP development); (5) Path: Specifies the location of the file on the server (for example, /pages/dashboard.html); (6) Query String: Key-value parameters passed to server scripts starting with '?' (such as ?role=student&status=active); and (7) Fragment: Starts with '#' to scroll directly to a specific HTML element ID.

2. How does an HTML file get processed in a web browser?

When a browser receives an HTML document, it follows these rendering steps: (1) Byte Conversion: Raw network bytes are decoded into characters based on the UTF-8 charset; (2) Tokenization: Characters are parsed into HTML start tags, end tags, attributes, and text; (3) DOM Tree Construction: Tokens form the hierarchical Document Object Model (DOM); (4) CSSOM Construction: External and embedded CSS is parsed into the CSS Object Model; (5) Render Tree: Visible DOM elements are merged with computed styles; (6) Layout: The browser computes the exact dimensions and position of each element; and (7) Painting: Pixels are drawn on the screen to display the webpage.

3. How will page navigation flow be managed among all HTML pages?

Navigation is managed using a consistent top navbar across all pages with relative pathing. Public visitors can view informational pages (Home, About, Events, Contact, Login, and Register). Once authenticated, students are routed to their personal dashboard, while administrators are directed to the admin management modules. Direct access to protected internal pages checks session state and redirects unauthenticated users to the login page.

4. How will GitHub commits be maintained after each practical?

Git version control is maintained at https://github.com/neel92654/WDF_A_ARP. Each practical milestone is committed with descriptive messages documenting requirement analysis, static templates, responsive layouts, client validation, and backend PHP/MySQL features. A root .gitignore file excludes system files and cache, while the repository README provides setup instructions and project module details.

B. Implementation Details

I initiated the Student Hub project by identifying the requirements for CHARUSAT engineering students and faculty. I planned a core set of 10+ interconnected pages (Home, About, Register, Login, Dashboard, Attendance, Courses, Timetable, Profile, Events, Contact, Admin Dashboard, and Student Management). I used the project sitemap and low-fidelity wireframes from my GitHub repository (https://github.com/neel92654/WDF_A_ARP) to organize the navigation structure, and set up a modular project folder layout separating css/, js/, pages/, php/, sql/, storage/, and docs/.

C. Evidence of Work

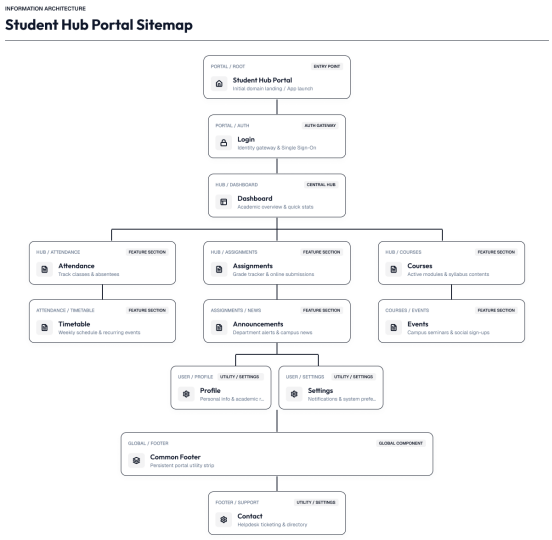


Figure 1: Actual Project Sitemap from GitHub Repository (docs/SiteMap.png)

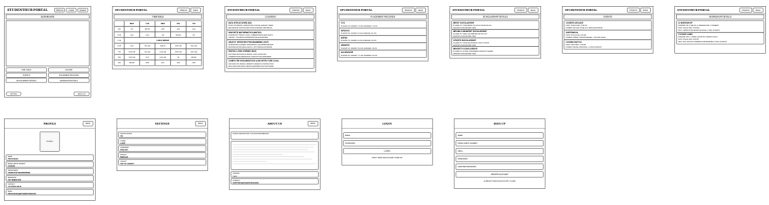


Figure 2: Actual Low-Fidelity Wireframe from GitHub Repository (docs/P1Wireframes.svg)

D. Repository Link

GitHub Repository: https://github.com/neel92654/WDF_A_ARP

E. Learning Outcomes

I learned how to conduct software requirement analysis, design user navigation sitemaps, construct low-fidelity wireframes, establish a clean directory structure, and organize milestone-based version control for a multi-page web application.